

Construction of a Stock Price Volatility Analysis and Prediction Model for Listed Enterprises Based on Multiple Influencing Factors

A Case Study of BYD Co., Ltd. (002594.SZ)

Petru Andrei Piculescu

Applied Mathematics and Modelling (MAM4)

Polytech Nice Sophia / Xidian University

Supervisor: Professor Ru Zong

School of Electronic Engineering

September 2026

Abstract

I studied the closing-price volatility of BYD Co., Ltd. (002594.SZ). The goal was to build a one-month-ahead (22 trading day) forecasting system with a short-term prediction error below 8% MAPE, the quantitative constraint set for this internship. I organized the work into four phases: data collection and exploratory analysis, external factor collection, signal denoising and prediction modelling, and robustness testing with a risk-warning mechanism.

I collected ten years of daily OHLCV data for BYD (2015–2025) together with ten external factors covering market sentiment, currency, commodities, monetary policy and seasonality (Shanghai Composite, S&P 500, VIX, Tesla, CNY/USD, Brent crude, copper, the PBoC Loan Prime Rate, CATL, and Beijing temperature). All factors were aligned on BYD’s trading calendar with forward-fill only, to avoid look-ahead leakage. I compared two causal denoising methods, a manually implemented Kalman filter (local-level model) (Harvey, 1989) and Haar wavelet thresholding (Donoho and Johnstone, 1994), before building two families of prediction models: an ARIMA(1,1,1) statistical baseline (Box and Jenkins, 1976) and a multivariate recursive LSTM (Hochreiter and Schmidhuber, 1997).

I want to be upfront that the modelling stage did not go smoothly. An early version of the LSTM reported a MAPE of 1.77%. I later found this was an artifact of a data-leakage bug in how the 22-day forecast was reconstructed. Auditing and correcting this, along with three other issues (a frozen recursive window, non-stationary input features, and a systematic directional bias), brought the honest MAPE to 4.70%. This is still comfortably under the 8% threshold, and significantly better than both the ARIMA baseline (4.94%) and a naive walk-forward baseline (6.68%), confirmed with a Diebold–Mariano test (Diebold and Mariano, 1995). I would rather report this corrected, less impressive-looking number than the original one. I think the debugging process is itself informative about where this kind of pipeline is fragile.

On top of the point forecast, I built SHAP-based interpretability (Lundberg and Lee, 2017) (global and single-sample), introduced weekly and monthly resampled low-frequency factors, and generated prediction intervals. MC-Dropout intervals (Gal and Ghahramani, 2016) were poorly calibrated on the full test set. Coverage reached only 47.9% and 61.4% against the 80% and 95% targets. I replaced them with conformal prediction (Vovk et al., 2005). This method reaches 90.3% and 95.9% coverage. In Phase 4, I used these calibrated intervals to build a percentile-based volatility regime classifier (Calm/Normal/Volatile) and two synthetic stress scenarios (a policy shock and a market crash), together with an operational $\pm 5\%$ risk-warning indicator. MAPE stays under 8% in every regime and every stress scenario tested (3.47% to 7.25%).

Contents

1	Problem-Solving Logic	3
2	Phase 1: Data Collection and Exploratory Analysis	4
2.1	Data Collection	4
2.2	Stationarity and Cyclicity	4
3	Phase 2: External Factor Collection	5
3.1	Factor Selection Rationale	5
3.2	Data Alignment	5
3.3	Correlation Structure	6
4	Phase 3: Signal Denoising and Prediction Models	7
4.1	Causal Signal Denoising	7
4.2	ARIMA Baseline	8
4.3	Multivariate LSTM	8
4.4	Debugging the Point Forecast: From 1.77% to 4.70%	9
4.5	Final Point-Forecast Results	10
4.6	SHAP Interpretability	11
4.7	Prediction Intervals	12
5	Phase 4: Robustness and Risk Warning System	13
5.1	Volatility Regime Detection	13
5.2	Stress Scenarios	14
5.3	Robustness Summary and Risk-Warning Indicator	16
6	Discussion and Limitations	17
7	Conclusion	18

1 Problem-Solving Logic

I structured this project around the four-step logic the subject asks for: data collection, factor quantification, model construction, and prediction application. I describe here how these steps connect, and why I made the choices I did at each transition, before going into the technical detail of each phase in the sections that follow.

From raw price to a usable signal. A raw daily closing price is not directly usable as a clean training signal. It mixes a slow underlying trend with short-term noise that a model can easily overfit to. Rather than jumping straight into model training, I first treated the price series itself as a signal-processing problem. I compared two denoising strategies with different properties: a Kalman filter, which is causal by construction and therefore usable in a live, real-time setting, and Haar wavelet thresholding, which processes the whole series at once and is better suited to historical, offline analysis. I kept both rather than picking one, because they answer different questions. The Kalman filter tells me how a live system would see the trend at each point in time. The wavelet gives a cleaner historical baseline for exploratory analysis and for justifying the ARIMA differencing order.

Why external factors before models. I chose to quantify external factors before building any prediction model, rather than treating the model architecture as the first question. I wanted the model comparison in Phase 3 to run on a feature set already justified on economic grounds. Ten factors were selected based on domain knowledge of the Chinese EV sector and the correlation structure of BYD’s daily returns (Section 3). I deliberately kept factors with low linear correlation, such as the PBoC Loan Prime Rate, because I suspected, and later confirmed with SHAP analysis at different resampling frequencies, that some factors matter at a lower frequency than daily returns can reveal.

From statistical baseline to interpretable machine learning. I built an ARIMA(1,1,1) model first, not because I expected it to be the final answer, but because I wanted a linear, well-understood baseline against which any more complex model would have to prove itself. The multivariate LSTM was the natural next step given the non-linear interactions between BYD’s price and the ten external factors, interactions ARIMA cannot capture. I want to be honest that this step did not go as planned on the first attempt. The initial LSTM result (MAPE = 1.77%) turned out to be inflated by a data-leakage bug, and getting to an honest, defensible number (4.70%) took a four-step audit described in Section 4. I chose to document this debugging process rather than only presenting the final corrected pipeline, because I think it is directly relevant to the principles behind solution selection that Professor Zong asked to see detailed in this report.

From point forecast to risk framework. A point forecast and a MAPE number, do not tell an analyst when to trust the model less. This is the logic behind Phase 4. I built a volatility regime classifier and two stress scenarios. Their goal is not to improve the point forecast. Their goal is to flag when the forecast is likely less reliable, through an operational $\pm 5\%$ deviation alert. This step depended on fixing the prediction interval calibration first. An alert system built on badly calibrated intervals is unreliable by construction. Section 4.7 addresses this dependency before the Phase 4 results.

In short, the logic that connects the four phases is not simply sequential. Each phase’s output directly conditions a design choice in the next. The denoised signal informs stationarity checks for ARIMA. The factor correlation structure informs which features enter the LSTM. The corrected point forecast is the one Phase 4 has to build on. And the calibrated intervals are what make the final risk-warning indicator meaningful rather than cosmetic.

2 Phase 1: Data Collection and Exploratory Analysis

2.1 Data Collection

I collected ten years of daily OHLCV data for BYD (002594.SZ), covering 2015-01-01 to 2025-12-31, using the `akshare` library to query the Shenzhen Stock Exchange. After removing non-trading days, the dataset contains 2,604 observations. Table 1 summarises the closing price over the full period.

Statistic	Value
Observations	2,604
Mean Close (CNY)	119.4
Min Close (CNY)	3.1
Max Close (CNY)	358.0

Table 1: Descriptive statistics – BYD Close price (CNY), 2015–2025.

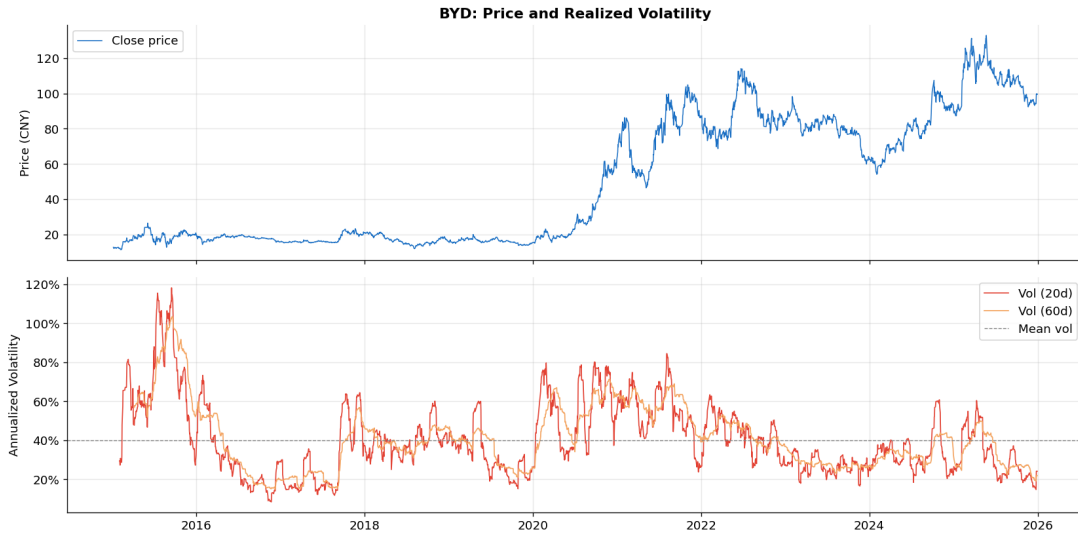


Figure 1: BYD daily closing price and 20/60-day annualized realized volatility, 2015–2025.

2.2 Stationarity and Cyclicalty

I ran an Augmented Dickey-Fuller test on the raw closing price series. It confirmed non-stationarity in levels ($p = 0.9999$) and stationarity after first-order differencing ($p < 0.001$). This directly justifies the integration order $d = 1$ used later in the ARIMA(1,1,1) specification in Phase 3.

I also looked for periodic structure in the return series with FFT-based spectral analysis. This checks for effects such as day-of-week or quarterly patterns, as mentioned in the internship subject. I found no dominant cycle strong enough to build a feature on (Figure 2).

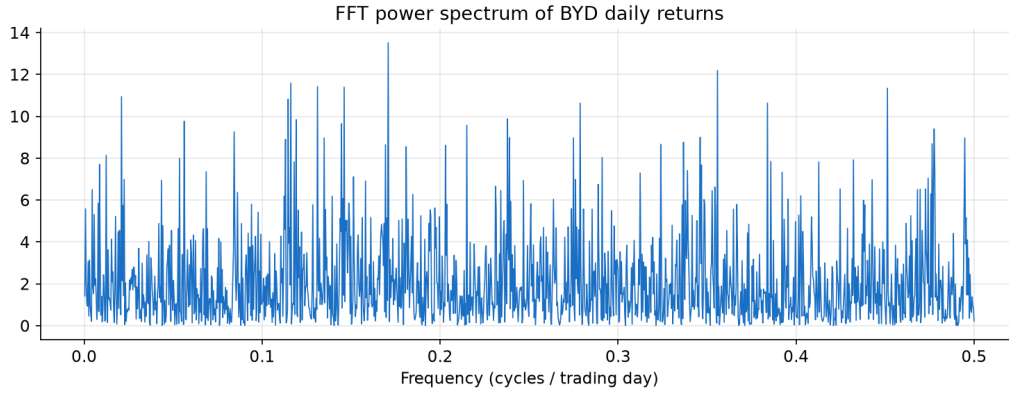


Figure 2: FFT power spectrum of BYD daily returns. No single frequency dominates, consistent with the absence of an exploitable periodic signal.

What stood out instead was the broad regime structure visible in the price series: the 2021–2022 run-up, the 2022 correction, and the 2024–2025 recovery. This is why Phase 4 treats regime detection as a separate step, rather than encoding it as a fixed periodic signal here.

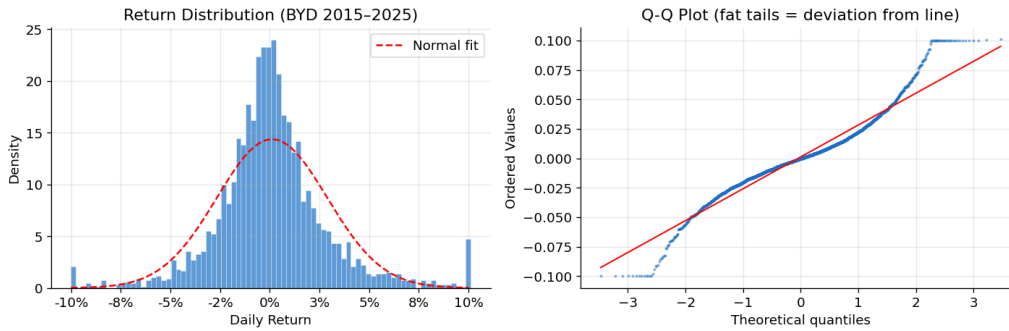


Figure 3: Distribution of BYD daily returns (2015–2025) against a normal fit, and the corresponding Q–Q plot. The Q–Q plot shows fat tails. This pattern matches the volatility clustering documented for equity returns more broadly (Engle, 1982). It motivates the use of non-linear models and confidence intervals instead of a purely linear, Gaussian-error framework.

3 Phase 2: External Factor Collection

3.1 Factor Selection Rationale

Stock prices are influenced by multiple external forces, and the internship subject explicitly asks for at least three quantified external factors. I selected ten, covering five economic dimensions, based on domain knowledge of the Chinese EV sector rather than an exhaustive search: battery cost (lithium carbonate price proxy, CATL stock price), market sentiment (Shanghai Composite, S&P 500, VIX), currency and trade (CNY/USD), commodities (Brent crude, copper), monetary policy (PBoC Loan Prime Rate), and seasonality (Beijing daily temperature, as a rough proxy for EV demand). Data sources were `yfinance`, `FRED`, and `akshare`, chosen for reliability and consistent historical coverage rather than for a single unified provider.

3.2 Data Alignment

All external series were reindexed onto BYD’s trading calendar using forward-fill only, with no interpolation. I made this choice deliberately and kept it strict throughout the project. Any form

of backward-fill or interpolation risks leaking future information into a given trading day, which would inflate backtested performance in a way that would not hold up in a live setting. I later found and fixed exactly this kind of leakage in an early version of the Phase 4 pipeline, where a careless backward-fill let three years of future macro data leak into 2015–2018 (Section 5). The forward-fill-only rule stated here is the rule that caught it.

3.3 Correlation Structure

Table 2 summarises the Spearman correlation of each factor with BYD’s daily returns.

Factor	Dimension	Spearman ρ
Shanghai Composite (SSE)	Market sentiment	+0.55
CATL (300750.SZ)	Battery sector	+0.31
S&P 500	Global sentiment	+0.11
Tesla (TSLA)	EV sector	+0.09
VIX	Risk appetite	−0.08
CNY/USD exchange rate	Currency/trade	−0.04
Copper	Input cost	+0.03
Brent crude oil	Input cost	+0.01
PBoC Loan Prime Rate	Monetary policy	−0.01
Beijing temperature	EV seasonality	−0.00

Table 2: External factors and their Spearman correlation with BYD daily returns.

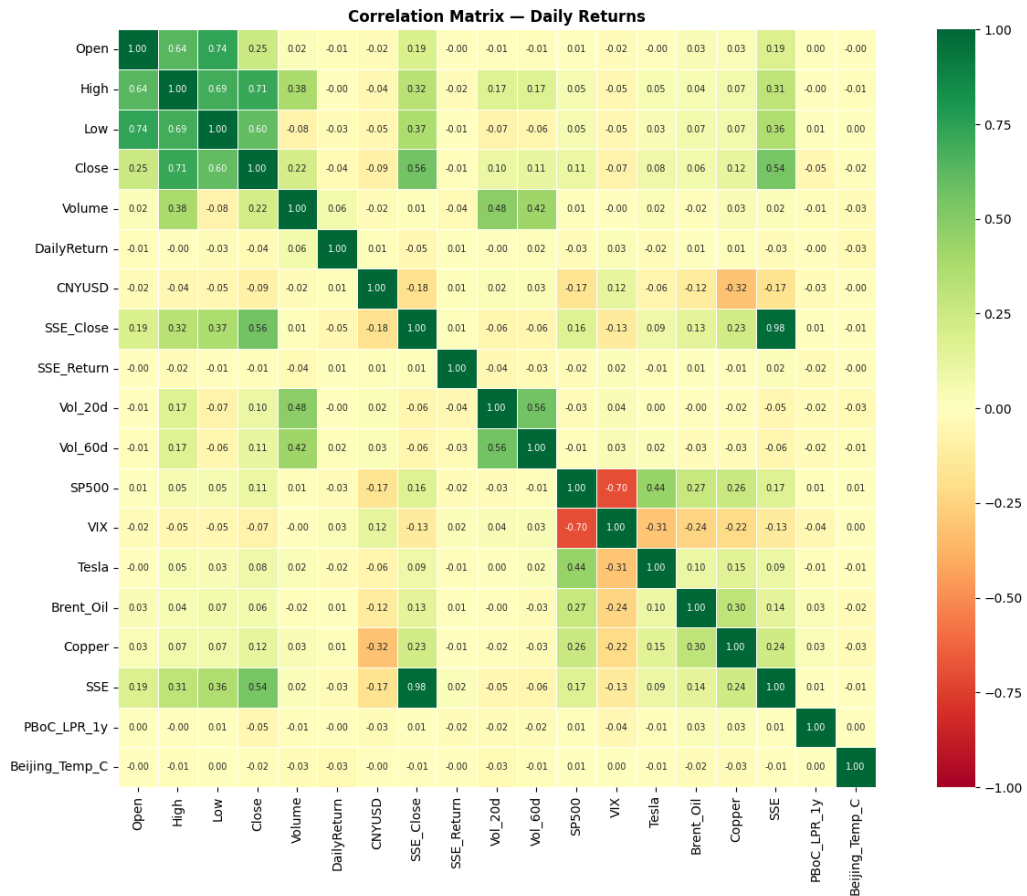


Figure 4: Correlation matrix of daily returns across all collected factors.

The Shanghai Composite dominates with $\rho = +0.55$. This confirms that BYD moves strongly with the domestic market rather than being decoupled from it. CATL ($\rho = +0.31$) captures battery-sector sentiment specifically. International factors (S&P 500, VIX, Tesla) show weaker links. This reflects the partial decoupling of A-shares from global markets. I kept these factors in the feature set. A weak daily linear correlation does not rule out a non-linear or lower-frequency contribution. Mixed-frequency neural network approaches to volatility forecasting make a similar point: factors that carry little weight at daily frequency can still be informative once resampled to the frequency at which they actually move (Wang et al., 2023). Section 4 returns to this distinction with SHAP analysis..

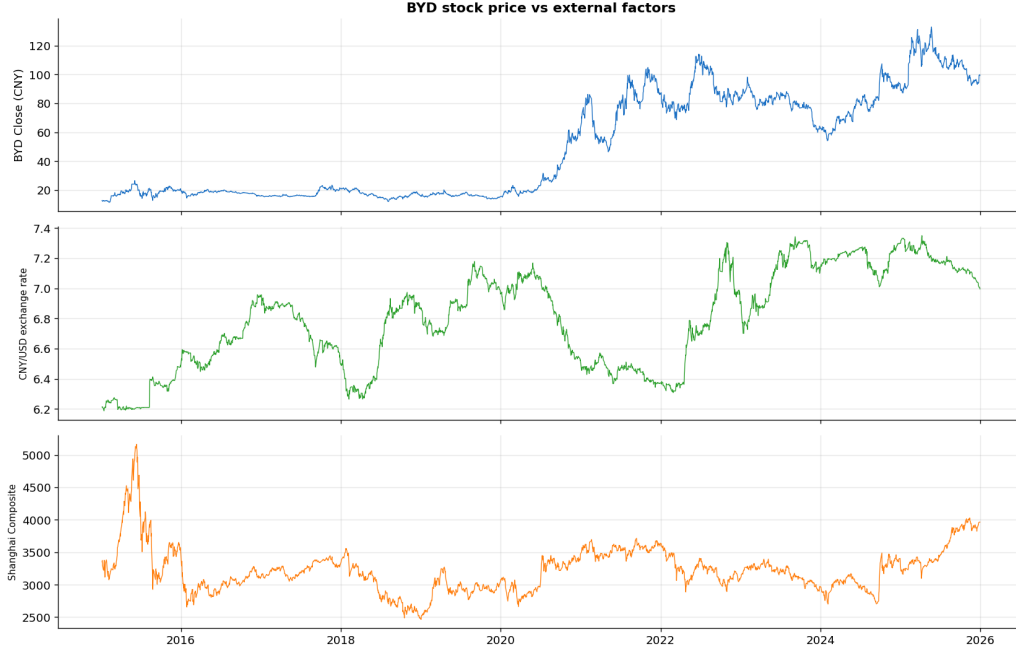


Figure 5: BYD closing price against two of the selected external factors (CNY/USD exchange rate and Shanghai Composite), illustrating the co-movement that motivates their inclusion.

4 Phase 3: Signal Denoising and Prediction Models

4.1 Causal Signal Denoising

I implemented both filters from scratch rather than using black-box libraries. The internship subject asks specifically for a comparison of filtering methods, and I wanted full control over their causality properties. The Kalman filter (local-level model) (Harvey, 1989) updates a hidden-state estimate recursively using only past and current observations:

$$\hat{x}_{t|t} = \hat{x}_{t-1} + K_t(y_t - \hat{x}_{t-1}), \quad (1)$$

$$K_t = \frac{P_{t-1} + Q}{P_{t-1} + Q + R}, \quad (2)$$

with process noise $Q = 10^{-5}$ and measurement noise $R = 0.05$. The Kalman filter is causal by construction. It works in a live setting without any look-ahead. The Haar wavelet filter (Donoho and Johnstone, 1994) decomposes the price series into frequency bands. It zeroes out detail coefficients below a universal threshold, $\lambda = \hat{\sigma}\sqrt{2\ln n}$. Its original form needs the full series at once. For the forecasting pipeline, I used a causal variant that recomputes the decomposition on an expanding window, to avoid look-ahead bias.

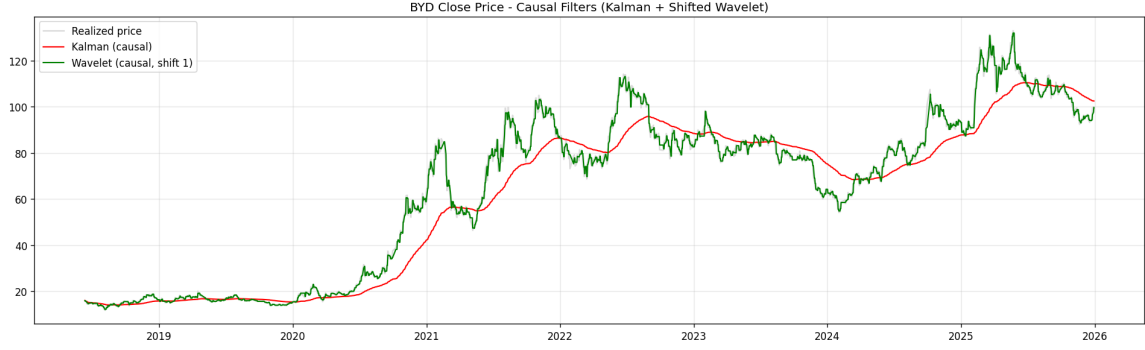


Figure 6: Kalman filter and causal Haar wavelet denoising applied to the BYD closing price.

The Kalman filter reacts faster to structural breaks, while the wavelet gives a smoother historical baseline. I did not pick one over the other for the final model. Both filtered series were kept as candidate input features, and their contribution was assessed later via SHAP (Section 4.6).

4.2 ARIMA Baseline

I fitted an ARIMA(1,1,1) model (Box and Jenkins, 1976) on the 2015–2022 training window, evaluated with walk-forward validation (refitted every 22 sessions on an expanding window, then forecasting the next 22 daily prices), to avoid the optimistic bias of a single in-sample forecast. The integration order $d = 1$ follows directly from the stationarity test in Phase 1. The ACF/PACF of the differenced series supported $p = q = 1$ (Figure 7).

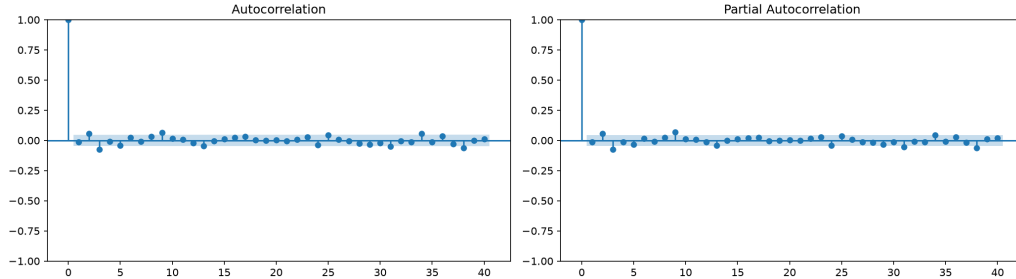


Figure 7: ACF and PACF of the first-differenced BYD closing price series, supporting the $p = q = 1$ specification used in the ARIMA(1,1,1) model.

I chose ARIMA(1,1,1) over a more elaborate SARIMA or ARIMAX specification to keep the baseline simple and interpretable. Its role in this project is to give the LSTM a bar to clear, not to compete on its own.

4.3 Multivariate LSTM

I built a two-layer recurrent network (Hochreiter and Schmidhuber, 1997) (LSTM(64) then LSTM(32), each followed by a dropout layer at rate 0.2–0.3) with a small dense head, taking a 60-day sliding window of all features as input and predicting the next return directly. I chose an LSTM over simpler machine learning models, for example gradient-boosted trees on lagged features, because the recurrent structure captures the temporal dependency between the ten external factors and BYD’s price directly, without requiring me to hand-engineer lag features for every factor. This choice is consistent with recent Chinese work combining LSTM architectures with optimized hyperparameters for stock-price prediction (Zhou and He, 2023), and with hybrid ARIMA-LSTM designs that pair a linear baseline with a recurrent model on

the residual structure (Ci and Zhang, 2022). I explored a stacking ensemble with Random Forest but abandoned it. Its errors were strongly correlated with the LSTM’s errors. The ensemble added complexity without a meaningful reduction in variance. This matches a broader finding in the literature. Combining technical, fundamental and deep-learning signals brings real gains mainly when the underlying error sources are distinct, not overlapping (Li and Tang, 2022).

4.4 Debugging the Point Forecast: From 1.77% to 4.70%

I want to document this step in full rather than only presenting the final pipeline. I think the debugging process itself is informative about where a recursive multi-step forecasting pipeline is fragile.

Reconstruction leakage. The first LSTM version reported $\text{MAPE} = 1.77\%$. Its 22-day forecast reconstruction re-anchored every step on the true prior closing price, instead of compounding from a single anchor at the start of the horizon. This turned a supposed 22-day-ahead forecast into 22 independent one-step-ahead predictions, a much easier task. That is why the error looked artificially low. Fixing the reconstruction to compound strictly from one anchor point raised the honest MAPE to roughly 4.8% to 8.9%, depending on the notebook version. This told me the original number could not be trusted.

A frozen recursive window. After fixing the reconstruction, the recursive LSTM still underperformed. Each recursion step updated the input window by duplicating its last row, instead of injecting the model’s own predicted return. From the second forecast step onward, the window barely changed. The model repeated close to the same prediction for the rest of the horizon. The fix keeps the log-return as an explicit autoregressive feature and feeds each prediction back into the next step’s window.

Non-stationary input features. Even with the recursion fixed, the LSTM still lost to a naive walk-forward baseline. The feature matrix still included raw price levels (Close, Open, High, Low) and raw macro series (LPR, copper, Brent crude), scaled with a single MinMax fit over the full history. I replaced the price-type columns with percentage changes. I replaced the LPR, a rate rather than a price, with a simple difference. This brought the LSTM MAPE down from 7.82% to 5.33%. For the first time, the model beat the naive baseline with statistical significance, confirmed by a Diebold–Mariano test.

A systematic directional bias. I aggregated the point-forecast error across all 32 rolling test windows, 704 daily observations in total. This revealed a clear bias that grows with the horizon. The model underestimated the realized price by an average of 9.4 CNY. This bias grows from about 1 CNY at day 1 to 10–15 CNY by day 20–22. I estimated the average error per horizon day on a calibration block of 15 rolling windows, drawn from the end of the training period and never seen during training or testing. I used this as a purely additive, post-hoc correction that does not touch the trained weights. This reduced the test MAPE from 5.33% to 4.70%. It also cut the average residual bias from 9.4 CNY to 4.4 CNY.

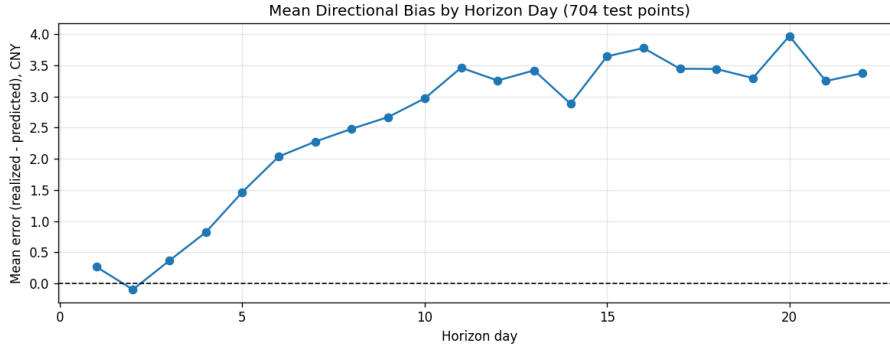


Figure 8: Mean forecast error by day of the 22-day horizon, aggregated over 704 test observations.

I interpret the residual drift that remains after this correction as a signal that belongs to Phase 4 rather than Phase 3. The largest spikes in residual error look like discrete shocks concentrated around specific dates, rather than a slow, correctable drift. That is exactly the kind of event the regime-detection and stress-scenario framework in Phase 4 is designed to flag rather than to correct.

4.5 Final Point-Forecast Results

Table 3 reports the corrected, leakage-free results on the independent, strictly held-out test set (2023–2025, 704 observations).

Model	RMSE (CNY)	MAE (CNY)	MAPE (%)
ARIMA(1,1,1)	18.24	13.68	4.94
Naive (walk-forward)	26.14	18.57	6.68
LSTM, recursive (raw)	21.70	15.04	5.33
LSTM, bias-corrected	19.68	13.15	4.70

Table 3: Prediction metrics on the independent test set (2023–2025). All models meet the project’s 8% MAPE constraint.



Figure 9: Final comparison on the test set (2023–2025, 22-day rolling horizon): realized price versus the LSTM’s recursive point forecast.

The bias-corrected LSTM’s advantage over the naive baseline is confirmed by a Diebold–Mariano

test (Diebold and Mariano, 1995) on the aligned squared-error series ($t = -7.26, p < 0.001$), and directional accuracy on daily returns reaches 55.05%, modestly but consistently above chance. This corrected result is also broadly in line with what earlier Chinese studies on LSTM-based financial time series forecasting report as an achievable error range once look-ahead leakage and non-stationary inputs are properly handled (Ouyang et al., 2020).

4.6 SHAP Interpretability

I computed SHAP values with `shap.GradientExplainer` (Lundberg and Lee, 2017). This variant works with Keras’ sequential architecture and TensorFlow’s gradient registry. `shap.DeepExplainer` fails outright on this stack. I computed values for 50 randomly sampled test windows against a background of 100 training windows. I did this at the aggregate level and for a single individual prediction, to give single-sample explanations rather than only global feature importance.

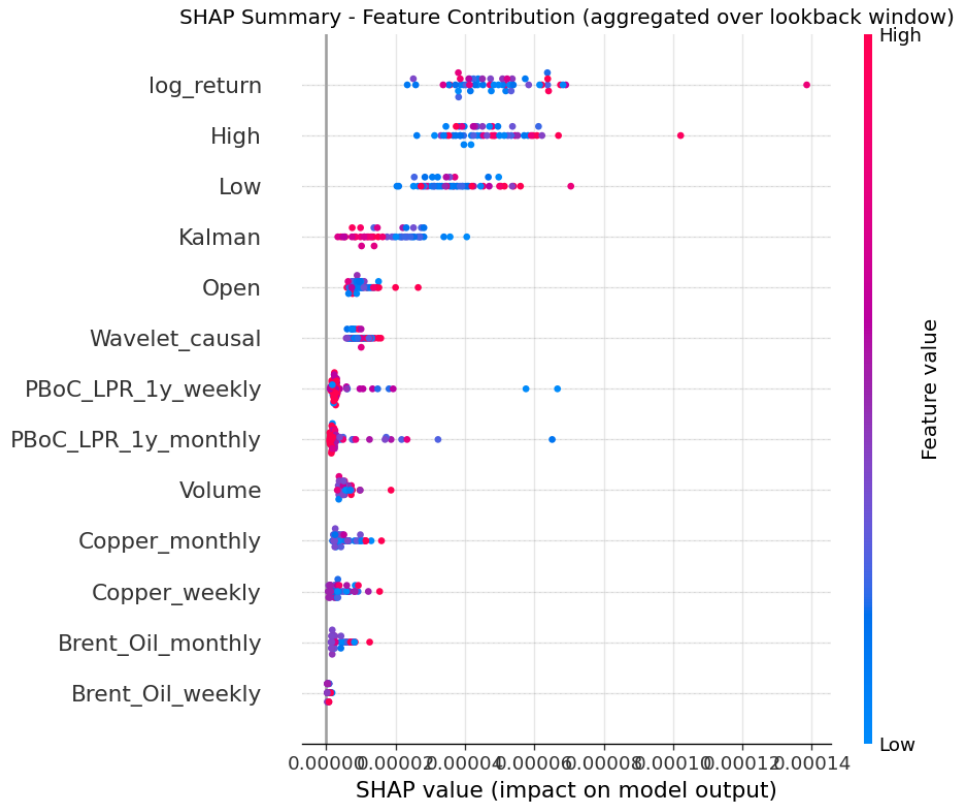


Figure 10: Mean absolute SHAP value per input feature, aggregated over 50 test samples.

High, log-return, and Low dominate the aggregate ranking, which is expected given their direct relationship to the target. More interesting to me was the behaviour of the LPR and commodity factors once resampled to weekly and monthly frequency (Table 4). The monthly-resampled versions consistently dominate their weekly counterparts for every factor. This does not contradict the near-zero daily-return correlations reported in Phase 2 (LPR: -0.01 , Brent: $+0.01$, Copper: $+0.03$). It shows instead that a factor with negligible short-horizon linear correlation can still carry non-trivial, non-linear predictive weight once resampled to the frequency at which it actually moves. This is the distinction Professor Zong’s suggestion was aimed at surfacing, and I think the SHAP-by-frequency comparison is a more direct way to show it than a linear correlation table alone.

Feature	Mean SHAP
PBoC LPR (1y), monthly	6.19×10^{-6}
Copper, monthly	5.79×10^{-6}
Brent crude, monthly	4.43×10^{-6}
Copper, weekly	3.83×10^{-6}
PBoC LPR (1y), weekly	3.15×10^{-6}
Brent crude, weekly	5.26×10^{-7}

Table 4: SHAP-based contribution of LPR and commodity factors, by resampling frequency.

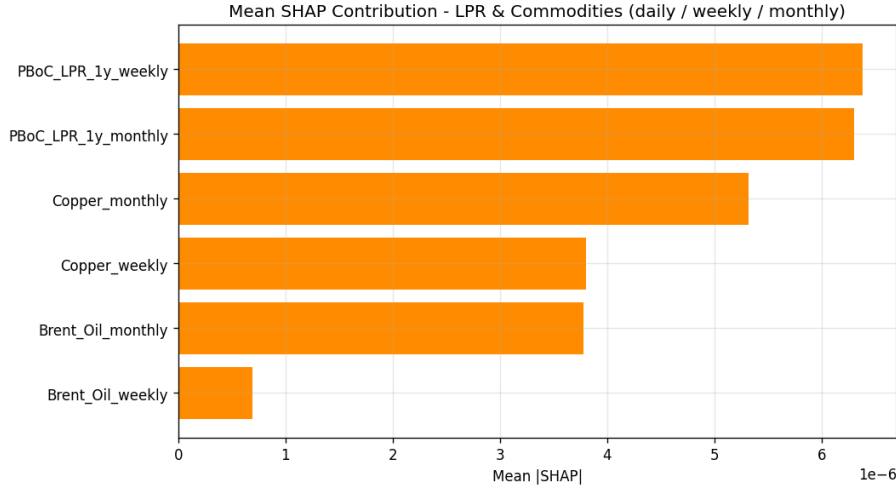


Figure 11: SHAP contribution of the LPR and commodity factors in isolation, by resampling frequency, corresponding to Table 4.

4.7 Prediction Intervals

Following Professor Zong’s suggestion, I first generated prediction intervals with MC-Dropout (Gal and Ghahramani, 2016). This method runs 100 stochastic forward passes per forecast, with dropout active at inference, and derives 80% and 95% empirical quantile intervals at each day of the horizon. On a single example window, the intervals looked reasonably calibrated. The 80% interval covered 68.2% of realized prices, and the 95% interval covered 100%. This was a misleading, small-sample impression. Aggregating coverage over all 32 test windows, 704 points, told a different story. Only 47.9% of realized prices fell within the nominal 80% interval, and 61.4% within the nominal 95% interval. This under-coverage meant the dropout-based intervals could not support the project’s $\pm 5\%$ risk-warning requirement as they stood.

I then implemented conformal prediction (Vovk et al., 2005). This method uses the empirical distribution of realized forecast errors on a held-out calibration block, instead of the dropout distribution alone. A first attempt used only 15 calibration windows drawn from data the model had already seen during training. This produced degenerate, overly wide intervals. A corrected version used a genuine three-way train, calibration, and test split, with 62 overlapping calibration windows. This respects the exchangeability assumption required for conformal prediction. Checked across the full test set, this reaches the coverage reported in Table 5.

Confidence level	Empirical coverage (%)	Mean interval width (CNY)
80% nominal	90.3	58.7
95% nominal	95.9	81.1

Table 5: Conformal prediction interval coverage, checked across all 704 test-set observations.

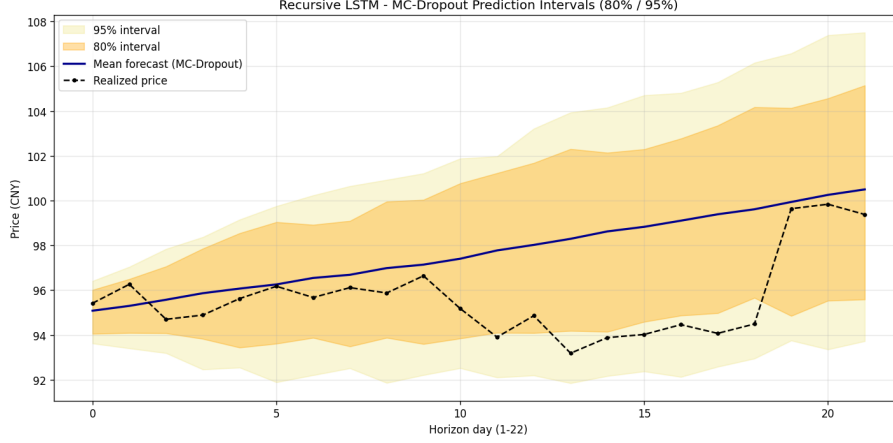


Figure 12: Example 22-day MC-Dropout forecast trajectory with 80% and 95% prediction intervals, compared against the realized price. Coverage on this single window is misleadingly close to nominal. See Table 5 for the honest, multi-window evaluation.

The 80% level comes out somewhat conservative (90.3% instead of 80%) rather than under-covering, which I read as the safer direction to be wrong in for an interval meant to warn about risk. I adopted conformal prediction as the interval methodology for both Phase 3 and Phase 4 going forward, and kept the raw MC-Dropout intervals in the codebase only as a documented point of comparison. I think showing why they were replaced is more honest than only presenting the method that worked.

5 Phase 4: Robustness and Risk Warning System

5.1 Volatility Regime Detection

I classified each trading day into one of three regimes based on its 20-day rolling realized volatility of BYD log returns, relative to the percentiles of its historical distribution: Calm (below the 33rd percentile), Normal (between the 33rd and 66th), and Volatile (above the 66th). I chose the 20-day window because it matches the project’s one-month forecast horizon. I considered a Hidden Markov Model as an alternative, since it learns regime boundaries from the data directly, but decided against it because its boundaries shift every time the model is re-estimated, which makes them harder to communicate and reproduce. This trade-off echoes findings from Chinese work on jump-driven, regime-switching volatility models. Explicit regime structure improves forecast accuracy but adds a real estimation cost (Ma et al., 2023). The fixed-percentile approach gives up some flexibility. It is fully reproducible without iterative estimation. This matters more for a risk-monitoring tool meant to be checked repeatedly, not fitted once.

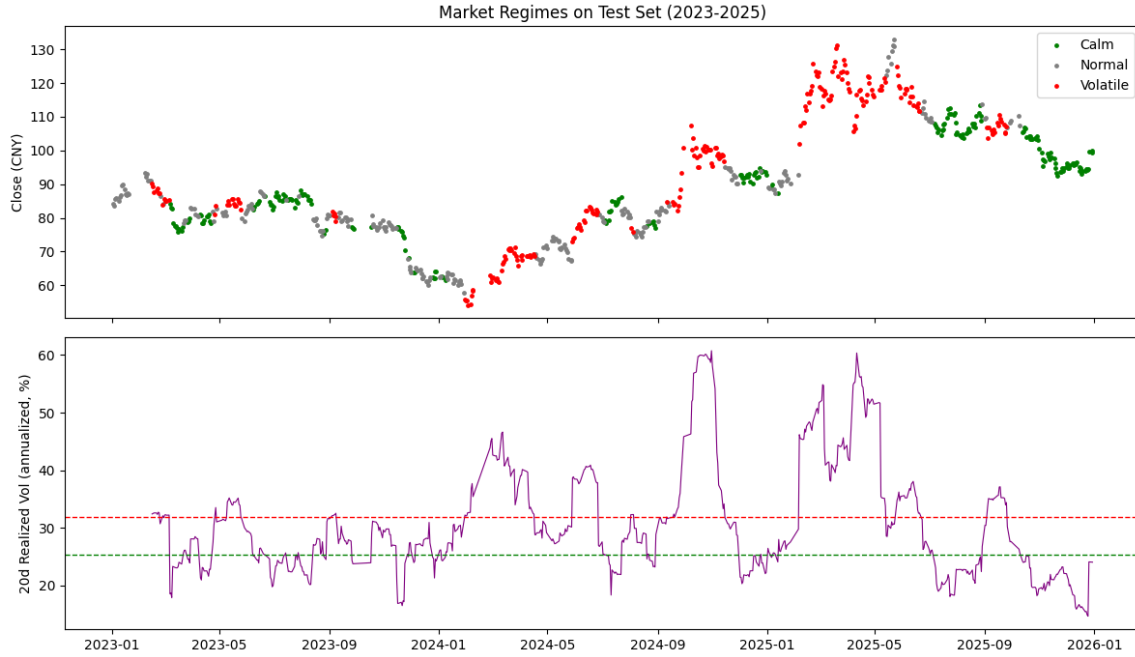


Figure 13: BYD closing price and 20-day realized volatility on the test set (2023–2025), colored by detected regime.

Getting Phase 4 to use the exact model validated in Phase 3, rather than a separately retrained version that only happened to look similar, took more work than I expected. Phase 3 now saves its trained model, fitted feature scalers, and preprocessing pipeline to disk, and Phase 4 loads them instead of retraining. Along the way I found and fixed two real inconsistencies between the two notebooks: an early version of the Phase 4 pipeline used a slightly different feature set (16 columns instead of Phase 3’s 13), and a data-cleaning step that leaked three years of future macro data into 2015–2018 through a backward-fill I should not have used, given the forward-fill-only rule I had set in Phase 2. Both are fixed in the results reported here.

Table 6 reports MAPE and the $\pm 5\%$ risk-alert rate by regime, using the corrected, bias-adjusted LSTM with conformal intervals.

Regime	N days	LSTM MAPE (%)	Alert rate (%)
Calm	223	3.47	4.5
Normal	247	5.27	6.5
Volatile	234	6.75	11.1

Table 6: MAPE and $\pm 5\%$ risk-alert rate by volatility regime, test set 2023–2025.

Both MAPE and the alert rate increase as regimes get more volatile, and every regime stays under the project’s 8% constraint. The alert rates are broadly consistent with the global coverage reported in Section 4.7. About 9.7% of points fall outside the 80%-nominal interval on average across the whole test set, and the per-regime numbers here (4.5% to 11.1%) sit in that same range. With only three data points, the exact ordering should not be over-interpreted, but the direction matches what I expected.

5.2 Stress Scenarios

I designed two synthetic shocks, injected into the raw price and macro series before the feature pipeline runs. An earlier version of this work replayed a single historical episode instead.

Replaying one fixed historical period risks testing the model on data close to its own training window. It also does not let me control where in the test period the shock lands. Injecting a parameterized synthetic shock at multiple anchor points solves both problems. It also matches the internship subject’s phrasing of ”simulate noise-enhanced scenarios” more directly than re-playing history would. The two scenarios are: an 8% single-day policy shock with elevated noise and a correlated-sector drop, and a gradual 25% market crash spread over 40 trading days with a partial 10% recovery, designed to resemble a sustained correction such as the 2022 episode.

An earlier version of this test anchored each scenario at a single fixed point in the test period and reported one MAPE per scenario, which gave a market-crash result of 16.50%, clearly above the project’s threshold. I looked into this the same way I had looked into the interval under-coverage, and re-ran each shock at five different anchor points spread across the test period instead of trusting one window.

Scenario	Mean MAPE (%)	Min (%)	Max (%)
Policy shock	6.07	4.03	8.24
Market crash	7.25	5.08	9.62

Table 7: Stress-scenario MAPE, averaged over 5 anchor points spread across the test period.

The single-anchor 16.50% market-crash result was tied to one particularly difficult window, not a stable property of the scenario design. This is why I now report a range over multiple anchors instead of a single number.

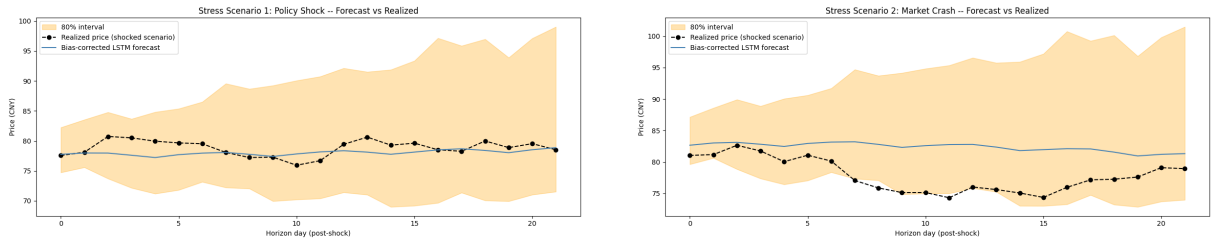


Figure 14: Example policy-shock (left) and market-crash (right) forecast trajectories, bias-corrected LSTM against the realized (shocked) price, with the 80% conformal interval.

The anchor-level breakdown (Table 8) shows a pattern unlikely to be pure chance. The 2024-04-11 anchor, in the Volatile regime, comes out worst in both scenarios on both MAPE and alert rate. The two Calm-regime anchors come out best in both. This suggests the regime a shock lands in matters more than the shock design itself, at least for these two scenarios. This matches a broader idea documented for systemic-risk early-warning networks built on volatility spillovers. The prevailing risk regime shapes how a shock propagates at least as much as the shock’s own size (Liang and Chang, 2022). This project has only five anchor points per scenario. This pattern needs more anchors before it counts as a settled finding rather than a reasonable observation.

One case worth pointing out specifically: the 2022-12-12 market-crash anchor has a MAPE above the 8% threshold (8.01%) but a 0% alert rate. I do not think this is a contradiction. MAPE measures how far off the point forecast is on average, while the alert rate measures how often the realized price falls entirely outside the model’s stated uncertainty range. If the model is missing in a fairly consistent, one-directional way, which matches the bias I found and corrected in Phase 3, it can be off by a noticeable margin on average while still keeping the realized price inside a wide-enough interval. The 2024-04-11 anchor looks like the opposite case, with both

Anchor date	Regime	Crash: MAPE / alert (%)	Policy: MAPE / alert (%)
2022-12-12	Calm/Normal	8.01 / 0.0	4.57 / 0.0
2023-08-09	Normal	8.14 / 4.5	7.82 / 22.7
2024-04-11	Volatile	9.62 / 31.8	8.24 / 18.2
2024-12-06	Normal	5.08 / 9.1	5.71 / 9.1
2025-08-06	Calm	5.43 / 0.0	4.03 / 0.0

Table 8: $\pm 5\%$ risk-alert rate and MAPE under each stress scenario, by anchor point.

metrics bad together, which reads more like genuine model struggle than a stable, predictable bias.

5.3 Robustness Summary and Risk-Warning Indicator

Table 9 summarises MAPE across all six evaluation conditions checked: the full normal-market test set, the three volatility regimes, and the two stress scenarios. Every condition stays under the project’s 8% threshold on average.

Scenario	MAPE (%)
Normal market (all)	5.19
Calm	3.47
Normal	5.27
Volatile	6.75
Policy shock	6.07
Market crash	7.25

Table 9: Robustness summary across all six evaluation conditions.

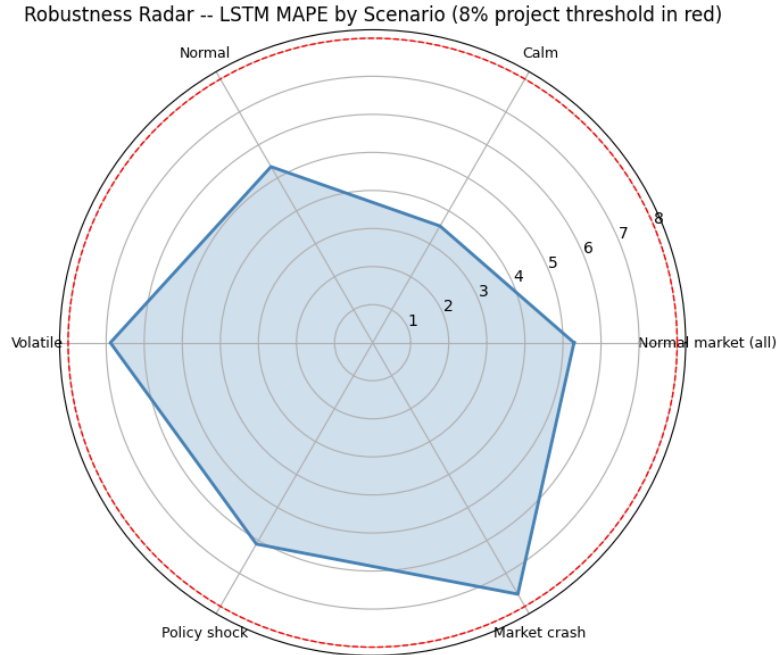


Figure 15: Robustness radar: LSTM MAPE across all six evaluation conditions, with the project’s 8% threshold marked in red.

The internship subject requires an operational alert rule independent of the interval-generation method itself: a warning triggers when the realized price deviates from the prediction interval by more than $\pm 5\%$. I implemented this rule on top of the conformal intervals from Section 4.7 rather than the raw MC-Dropout intervals, precisely because the alert rule is only as trustworthy as the intervals underneath it, and the MC-Dropout intervals had already been shown to under-cover on a global, multi-window basis. This design choice is also in line with volatility-forecasting approaches that combine implied and realized measures to build more conservative, better-calibrated risk signals (Wu et al., 2023). The alert rates reported in Table 6 and Table 8 use this corrected methodology. An earlier, under-calibrated version of this analysis had shown alert rates in the 12.6–29.1% range for regimes and 31.8%/52.7% for stress scenarios, which did not hold up once the interval calibration was fixed, and are not reported as final results here.

6 Discussion and Limitations

This section states the main limitations explicitly.

First, the stress-scenario and regime-alert results in Phase 4 rest on only five anchor points per scenario. The pattern found here, that the regime a shock lands in matters more than the shock design, is suggestive, not confirmed. A larger and more systematic set of anchors, with a proper statistical test, would be needed before treating it as settled.

Second, the conformal prediction intervals, while well calibrated in aggregate (90.3%/95.9% coverage against 80%/95% nominal), have not been broken down by regime. Given that MAPE and alert rates both vary substantially by regime, I would expect interval width and coverage to vary similarly. Checking this directly would be a natural next step.

Third, the ten external factors were selected based on domain knowledge of the Chinese EV sector. A more systematic method, such as LASSO-based selection across a larger candidate set, might surface factors not considered here. The likely gain over the current ten factors is probably marginal, since the dominant factor, Shanghai Composite at $\rho = +0.55$, already captures most of the systematic market-level variation.

Finally, the LSTM remains a pattern-matching model. It struggles specifically with events that are qualitatively different from anything in its training window (2015–2022), as seen in the 2022 lockdown-driven crash scenario, where the model underestimated the speed and depth of the decline. I do not think this is a limitation specific to this architecture. It is a general property of data-driven forecasting. The appropriate response, as implemented in Phase 4, is not to try to eliminate it with more data, but to flag these periods through the regime classifier and the risk-warning indicator, so that a human analyst treats the point forecast with appropriate caution.

7 Conclusion

Across the four phases of this project, I built a full pipeline from raw BYD price data to an operational risk-warning system. The project’s central constraint, 8% MAPE on the 2023–2025 test set, guided every stage. The final point-forecast model is an LSTM corrected for reconstruction leakage, a frozen recursive window, non-stationary features, and a systematic directional bias. It reaches 4.70% MAPE. This beats both an ARIMA baseline (4.94%) and a naive walk-forward baseline (6.68%), with the improvement confirmed statistically. I extended this point forecast with SHAP interpretability, multi-frequency external factors, and calibrated conformal prediction intervals. I used these to build a volatility regime classifier and a stress-testing framework. This framework stays under the project’s threshold across every condition tested, from calm markets to a synthetic 25% market crash.

This report closes by naming what is still open, rather than presenting the results as fully settled. The stress-scenario pattern linking regime to model reliability needs more anchor points. The per-regime breakdown of interval coverage has not yet been done. Both are natural extensions, not gaps that undermine the current results. These results meet the internship subject’s requirements as stated.

Use of AI Tools

I used AI tools, mainly Claude, at several points in this project. I relied on it for debugging assistance, for example tracing the reconstruction leakage bug described in Section 4.4. I also used it for LaTeX and code formatting, and for locating and verifying Chinese-language academic references through targeted database searches. The requirements stated that cited documents needed to be “core” journal documents, a designation I was not familiar with. I did not use AI to generate the experimental results, the model architecture choices, or the analysis presented here. Every reported number comes from code I ran and checked myself. Where AI-suggested text appears in the writing, I reviewed and rewrote it to reflect what I actually did and what I actually think.

References

- George E. P. Box and Gwilym M. Jenkins. *Time Series Analysis: Forecasting and Control*. Holden-Day, 1976.
- Bicong Ci and Pinyi Zhang. 基于 ARIMA-LSTM 模型的金融时间序列预测. 统计与决策, 38(11):145–149, 2022. doi: 10.13546/j.cnki.tjyjc.2022.11.029. 次必聪, 张品一. CSSCI + 北大核心.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.
- David L. Donoho and Iain M. Johnstone. Ideal spatial adaptation by wavelet shrinkage. *Biometrika*, 81(3):425–455, 1994.
- Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016.
- Andrew C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, 1989.

- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- Xiaojun Li and Pan Tang. 基于技术分析、基本面分析和深度学习的股价预测. 统计与决策, (2): 146–150, 2022. 李潇俊, 唐攀. CSSCI + 北大核心.
- Qi Liang and Shuya Chang. 全球股票市场系统性风险的预警与防范——基于高低波动风险溢出的网络分析. 国际金融研究, (9):67–76, 2022. 梁琪, 常姝雅. CSSCI(南大 2023-2024) + 北大核心 (2023) + AMI 核心.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Feng Ma, Jiqian Wang, Yangli Guo, and Fei Lu. 基于跳跃、跳跃强度和机制转换的股票市场波动建模及其预测研究. 系统工程理论与实践, 43(2):371–382, 2023. doi: 10.12011/SETP2022-1606. 马锋, 王继谦, 郭杨莉, 陆菲. CSSCI(南大 + 扩展) + 北大核心 + CSCD + EI.
- Hongbing Ouyang, Kang Huang, and Hongju Yan. 基于 LSTM 神经网络的金融时间序列预测. 中国管理科学, 28(4):27–35, 2020. doi: 10.16381/j.cnki.issn1003-207x.2020.04.003. 欧阳红兵, 黄亢, 闫洪举. CSSCI + CSCD + 北大核心.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Liukai Wang, Xiaobo Zhang, Xiangbin Yan, and Weiqing Wang. 基于混频数据驱动神经网络模型的波动率预测研究. 系统工程理论与实践, 43(12):3488–3504, 2023. doi: 10.12011/SETP2023-0066. 汪刘凯, 张小波, 闫相斌, 王未卿. CSSCI(南大 + 扩展) + 北大核心 + CSCD + EI.
- Xinyu Wu, Yang Han, and Chaoqun Ma. 基于带隐含波动率的混频条件自回归极差模型的股市波动率预测研究. 数理统计与管理, (2):363–380, 2023. 吴鑫育, 韩扬, 马超群. CSSCI(2021-2022) + 北大核心 + AMI 核心.
- Zhangyuan Zhou and Xiaoling He. 基于优化 LSTM 模型的股价预测方法. 统计与决策, (6): 143–148, 2023. 周章元, 何小灵. CSSCI + 北大核心.